

SPERMATOGENESIS

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- **Definition** : Process by which male gametes called **spermatozoa (sperms)** are formed from primitive spermatogenic cells (**spermatogonia**) in the testis.
- **Duration** : Takes **74 days** for the complete formation of a mature sperm from a primitive germ cell.
- **Sertoli Cell Attachment** : Spermatogenic cells maintain cytoplasmic connections with Sertoli cells, which supply essential nutrients and regulatory factors.
- **Stages** : Occurs in **4 sequential stages**:
 - a. Stage of Proliferation
 - b. Stage of Growth
 - c. Stage of Maturation
 - d. Stage of Transformation (*Spermiogenesis*)

SCHEMATIC OVERVIEW OF SPERMATOGENESIS

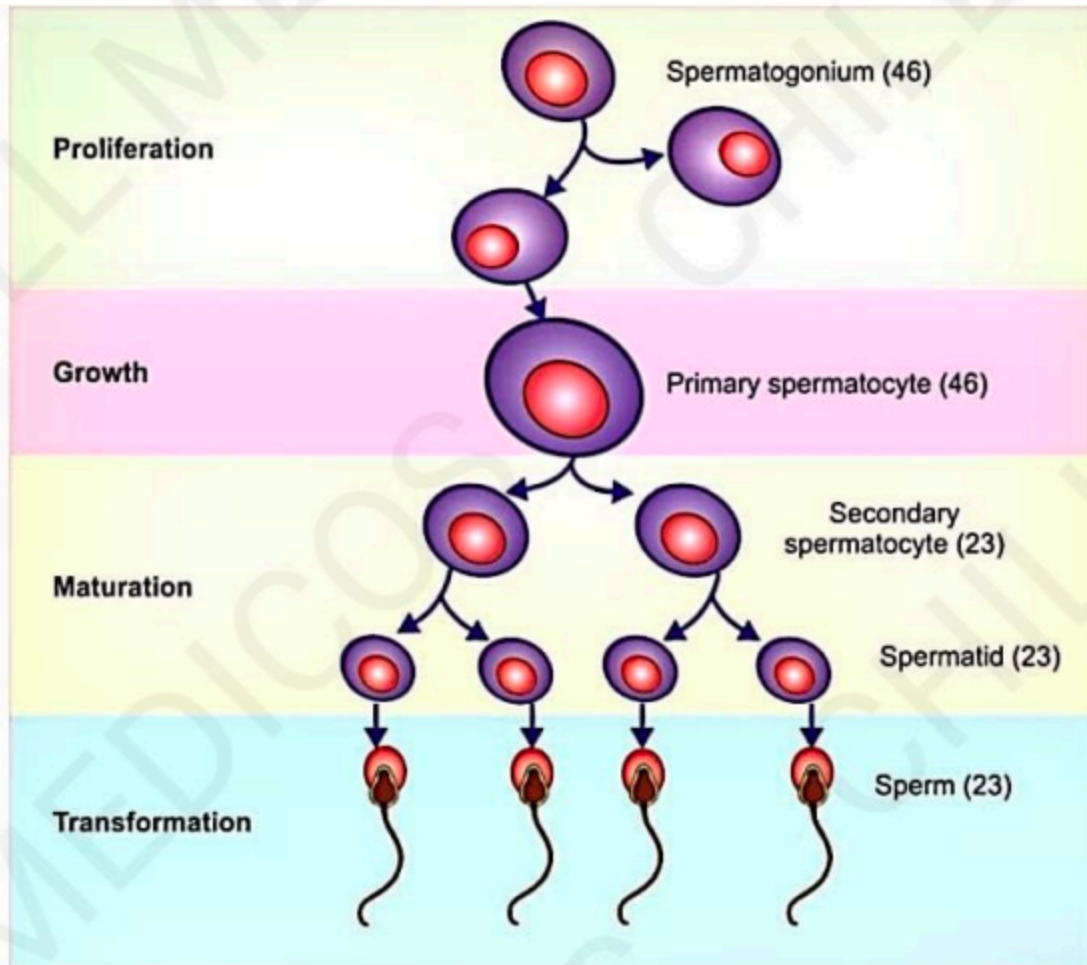


FIGURE 74.4: Spermatogenesis. Number in parenthesis indicate chromosomal number

1. Stage of Proliferation

Spermatogonium (46, 2n)

- 22 pairs autosomes + 1 pair sex chromosomes (XY)



Undergoes Mitotic
Divisions (7 generations in
humans)



2. Stage of Growth

Primary Spermatocyte (46, 2n)

- Grows into a large cell



3. Stage of Maturation (Meiosis)

1st Meiotic Division (Reductional)



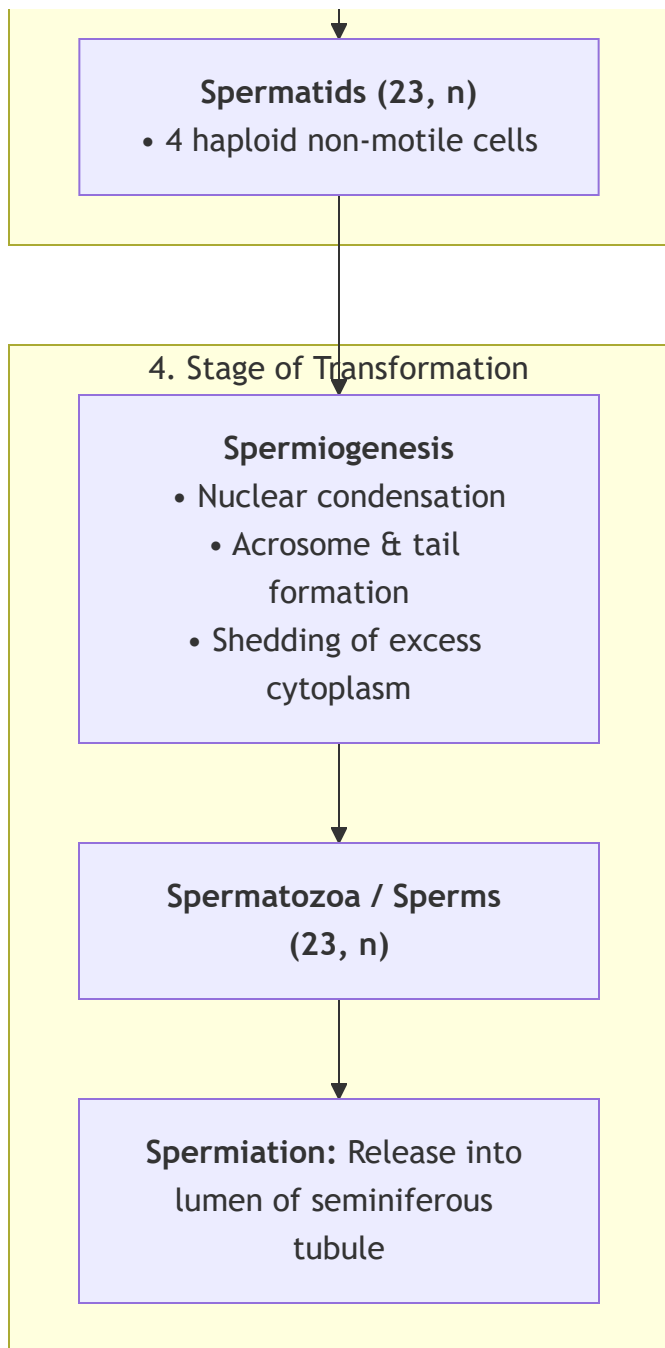
Secondary Spermatocytes (23, n)

- 2 haploid cells formed



2nd Meiotic Division (Equational)





DETAILED STAGES OF SPERMATOGENESIS

1. Stage of Proliferation

- Each spermatogonium contains **23 pairs of chromosomes** (22 pairs of autosomes + 1 pair of sex chromosomes $[XY]$).
- Spermatogonia divide repeatedly by **mitosis** without reducing chromosome number.
- In humans, spermatogonia undergo usually **7 generations** of mitotic division before the final generation enters the growth stage as a **primary spermatocyte**.

2. Stage of Growth

- The primary spermatocyte accumulates cytoplasm and grows into a **large cell** with no other major morphological changes.

3. Stage of Maturation (Meiotic Division)

- **First Phase (Meiosis I)** : Each primary spermatocyte (46 chromosomes) undergoes reduction division to form **2 secondary spermatocytes**, each receiving the **haploid number of chromosomes (23)**.
- **Second Phase (Meiosis II)** : Each secondary spermatocyte divides equationally into **2 smaller haploid spermatids (23 chromosomes)** (*yielding a total of 4 spermatids per primary spermatocyte*).

4. Stage of Transformation (Spermiogenesis & Spermiation)

- No further cell division occurs.
- **Spermiogenesis** : Morphological differentiation of round spermatids into streamlined, motile spermatozoa:
 - a. Condensation of nuclear chromatin.
 - b. Formation of the **acrosome**, middle piece (mitochondrial spiral sheath), and flagellar **tail**.
 - c. Removal/phagocytosis of extraneous, non-essential cytoplasm by Sertoli cells.
- **Spermiation** : The final detachment and release of mature spermatozoa from Sertoli cells into the **lumen of seminiferous tubules**.

REGULATION OF SPERMATOGENESIS

Spermatogenesis is influenced by:

- i. Sertoli cells
- ii. Hormones (*Hypothalamo-Pituitary-Gonadal Axis*)
- iii. Other physiological & environmental factors

1. Role of Sertoli Cells

- Support and nourish developing germ cells.

- Provide hormonal substances and paracrine regulators necessary for spermatogenesis.
- Secrete **Androgen-Binding Protein (ABP)** to maintain high local testosterone levels.
- Facilitate **spermiation** (release of sperms into tubular lumen).

2. Role of Hormones

- **Key Hormones Involved :—**
 - a. Follicle-Stimulating Hormone (FSH)
 - b. Testosterone
 - c. Estrogen
 - d. Luteinizing Hormone (LH)
 - e. Growth Hormone (GH)
 - f. Inhibin
 - g. Activin

Stage of Spermatogenesis	Hormones Necessary
Stage of proliferation	• Follicle-stimulating hormone (FSH) • Growth hormone (GH)
Stage of growth	• Testosterone • Growth hormone (GH)
Stage of maturation	• Testosterone • Growth hormone (GH)
Stage of transformation	• Testosterone • Estrogen

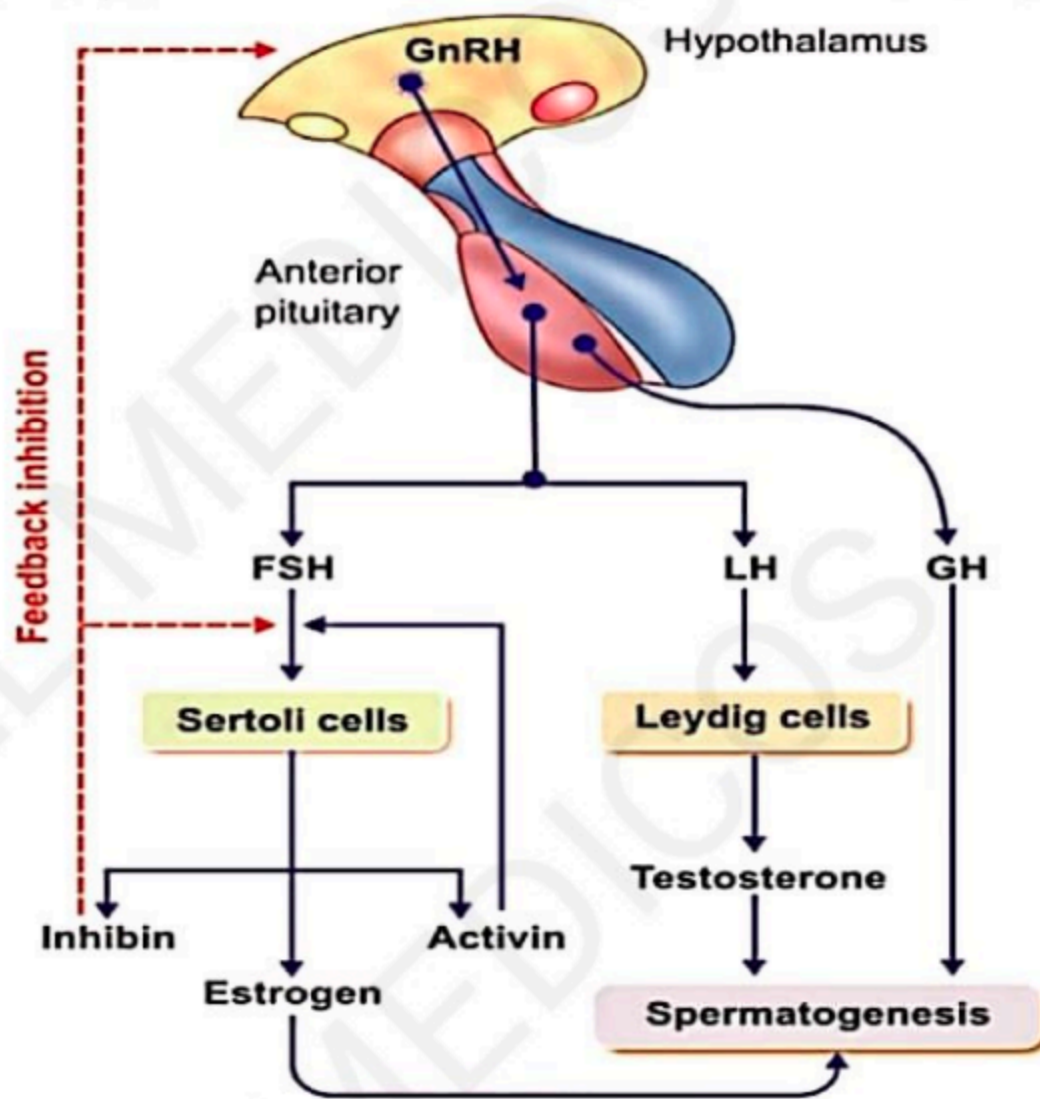
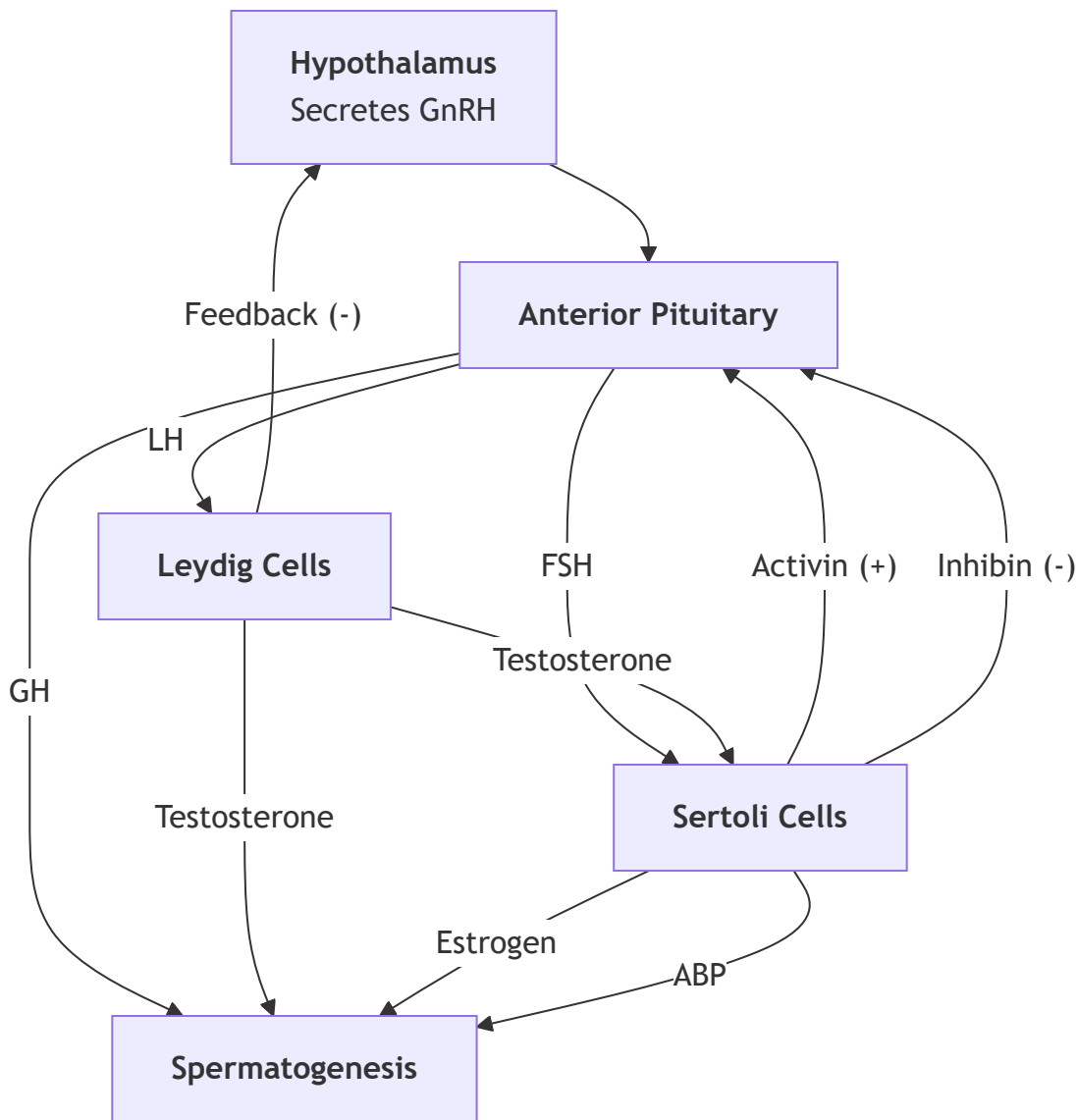


FIGURE 74.5: Role of hormones in spermatogenesis. Blue arrow = Stimulation, Red dotted arrow = inhibition, GnRH = Gonadotropin-releasing hormone, FSH = Follicle-stimulating hormone, LH = Lutinizing hormone, GH = Growth hormone.



Detailed Actions of Individual Hormones:

- **a. FSH (Follicle-Stimulating Hormone) :—**
 - Responsible for the **initiation of spermatogenesis**.
 - Binds to receptors on Sertoli cells and spermatogonia to induce mitotic proliferation.
 - Stimulates Sertoli cells to synthesize **Estrogen** and **Androgen-Binding Protein (ABP)**.
- **b. Testosterone :—**
 - Secreted by Leydig (interstitial) cells; responsible for the **maintenance of spermatogenesis** and progression of maturation stages.
 - Local concentration and activity are augmented by ABP.
- **c. Estrogen :—**
 - Converted from testosterone inside Sertoli cells via the enzyme aromatase; necessary for spermatid transformation.
- **d. LH (Luteinizing Hormone / ICSH) :—**

- In males, called **Interstitial Cell-Stimulating Hormone (ICSH)**.
- Essential for stimulating **Leydig cells to secrete testosterone**.
- **e. Growth Hormone (GH) :—**
 - Essential for baseline metabolic processes in the testes and promotes early proliferation of spermatogonia.
- **f. Inhibin :—**
 - Peptide hormone (transforming growth factor superfamily) secreted by Sertoli cells; selectively **inhibits FSH secretion** via negative feedback on the anterior pituitary.
- **g. Activin :—**
 - Peptide hormone secreted within gonads; ↑ **increases FSH secretion** and accelerates spermatogenesis.

3. Role of Other Factors

- **A. Temperature (Scrotal Thermoregulation) :—**
 - Spermatogenesis requires a temperature **2°C lower than core body temperature**, maintained by the scrotal sac.
 - Elevated testicular temperature stops spermatogenesis.
 - **Cryptorchidism (Undescended Testes) :** Testes remain in the abdomen → exposure to higher core body temperature → degeneration of seminiferous tubular epithelium → **complete arrest of spermatogenesis (azoospermia)**.
- **B. Infectious Diseases :—**
 - Systemic viral infections like **Mumps** (causing orchitis) can trigger inflammatory degeneration of seminiferous tubules, leading to temporary or permanent **stoppage of spermatogenesis**.